

Undesired corporate sustainability practices under imperfect ESG rating scheme

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Abstract

Purpose – This study examines the undesired impacts of one of China's ESG (Environmental, Social, and Governance) rating schemes on corporate sustainability practices in China. We draw on stakeholder salience theory and impression management theory to investigate how Chinese firms navigate ESG demands.

Design/methodology/approach – Using data from Chinese listed firms between 2011 and 2019, the study applies the 2015 enactment of China's New Environmental Protection Law as an exogenous shock to examine the investor and public concern about ESG ratings on firms' sustainability strategies.

Findings – The study finds that Chinese firms often prioritize symbolic sustainability initiatives over substantive environmental actions to enhance their ESG scores. This behavior is further exacerbated by the Chinese institutional context, where low social trust and high corruption fuel managerial opportunism and greenwashing.

Originality/value – Our findings highlight the need to strengthen environmental regulation, establish robust ESG rating schemes, and improve social oversight mechanisms to ensure that corporate practices contribute to environmental and social goals, rather than simply being used for image management.

Keywords ESG rating scheme, Environmental commitments, Greenwashing, Social trust, Corruption

Paper type Research article

1. Introduction

We must have zero tolerance for net-zero greenwashing. Today's Expert Group report is a how-to guide to ensure credible, accountable net-zero pledges, said António Guterres, Secretary-General of the United Nations, at the launch at COP27.

Investors and the general public are increasingly aware of the global trend towards corporate sustainability reporting standards, expecting firms to show the social and environmental benefits of their actions (Galbreath, 2013). Consequently, firms view green image as a competitive edge, while managers consider ESG (Environmental, Social, and Governance) ratings essential for boosting corporate reputation and competitiveness (Tan *et al.*, 2025). Yet, corporate ESG disclosures remain largely voluntary, with diverse reporting formats and lack of quantification—varying standards, low transparency, and poor comparability. The imperfect ESG rating scheme creates conditions that encourages opportunistic behavior among listed

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firms, turning ESG reports into instruments for corporate greenwashing and managerial career advancement. China, being the world's largest greenhouse gas emitter, has enacted numerous environmental laws and sustainable development reporting guidelines to promote a transition to a low-carbon economy, and ESG investments have gained significant traction (Yuan *et al.*, 2022). However, without stringent regulation and enforcement, the current reporting scheme [1] is often used by firms to polish their public image and by managers to advance their personal interests. Drawing on stakeholder salience theory and impression management theory, this study examines how the ESG rating scheme impacts the environmental strategies of Chinese listed firms.

Prior research highlights that firms use ESG disclosure strategically to enhance environmental reputation and legitimacy. For example, Walker and Wan (2012) find that firms under compliance pressure are more likely to disclose intended future actions—indicative of symbolic behavior—rather than current or completed initiatives that represent substantive environmental actions. Marquis *et al.* (2016) show that firms exaggerate environmental performance to raise valuations. Kirk and Vincent (2014) reveal that firms selectively highlight positive ESG outcomes while concealing negative ones, thereby misguiding stakeholders about true environmental performance. Studies also document sectoral variation in greenwashing. For instance, Herbohn *et al.* (2014) report that in the energy sector, environmental performance is positively associated with ESG disclosure, as firms prefer to disclose favorable environmental information to reinforce compliance. In contrast, Kim and Lyon (2015) find that U.S. electricity firms downplay environmental performance in disclosures to counteract the adverse impact of green certification on stock prices. Although these studies establish an initial link between ESG and greenwashing, existing approaches to identifying greenwashing rely heavily on ESG ratings and have yet to explain firms' strategic choice between symbolic and substantive environmental actions.

Firms engage with various stakeholders, including transactional ones like investors, employees, and suppliers, and non-transactional ones such as the natural environment, indigenous communities, and governments (Magness, 2008). These stakeholders not only contribute to corporate value but also serve as potential beneficiaries and risk bearers (Post *et al.*, 2002). Managers strive to balance the needs of all stakeholders to maximize corporate interests (Xue *et al.*, 2019). Nevertheless, stakeholder salience theory suggests they prioritize stakeholders based on power, legitimacy, and urgency, often neglecting secondary stakeholders due to limitations in time, attention, and resources (Mitchell *et al.*, 1997). As ESG ratings gain increasing attention from investors and the public, firms face the challenge of balancing stakeholder interests. Lanis and Richardson (2012) find that firms prioritize employees, customers, and communities over governments. Despite having high Corporate Social Responsibility (CSR) ratings, these firms may be less enthusiastic about government taxation. Conversely, Erdiaw-Kwasie *et al.* (2017) argue that firms often view communities as lower-priority stakeholders due to their limited influence and resources.

In the current ESG rating landscape, corporate ESG performance has become a tool for managers to address stakeholder expectations. Impression management theory explains how firms shape favorable impressions among investors and the public (Elsbach and Sutton, 1992). For instance, managers may introduce compensation policies that align with mainstream ideologies and institutional norms to strengthen organizational legitimacy. Alternatively, they may use media relations to present a socially responsible firm image, addressing stakeholders' concerns about internal governance (Tata and Prasad, 2015). As sustainable development principles gain traction, firms are using impression management not just to improve financial performance but also social responsibility performance. Managers emphasize positive environmental outcomes while downplaying or hiding negative ones to maintain a favorable image, garnering trust and expectations from stakeholders for economic gain (Tian *et al.*, 2024). As a result, firms focus on improving their ESG ratings to project a positive sustainability image, satisfying investors' ESG demands and complying with government regulations. However, stakeholders often lack access to complete information, making it hard to assess a

firm's true environmental impact behind its ESG scores. This leads firms to prioritize fulfilling ESG rating expectations from investors and governments rather than addressing the real environmental impact of their actions, factoring in the associated costs and risks.

As the world's largest emerging economy, China has advanced environmental legislation and rapidly developed its ESG framework in recent years (Chen *et al.*, 2025). The new *Environmental Protection Law*, enacted in January 2015 and widely known as the strictest in history, introduced key institutional changes. By shifting from fixed fines to daily penalties, imposing heavier sanctions on highly polluting sectors, and strengthening regulatory enforcement, it has pushed firms to place greater emphasis on ESG performance (Zhang and Han, 2025). Moreover, its provisions on environmental disclosure and public participation have enhanced institutional and investor oversight of both government and polluting firms (Wei *et al.*, 2025). Yet, environmental governance in China remains constrained by principal-agent challenges, as local governments lack incentives to prioritize environmental protection, hindering the effective execution of central government environmental goals (Wang and Ye, 2024). Given the lax enforcement of environmental regulations, China offers a unique context for examining how firms respond to rising investor attention to ESG ratings (Du, 2015). We select the enactment of China's 2015 *Environmental Protection Law* as an exogenous shock that intensified firms' and the public's focus on ESG ratings.

Using the Difference-in-Differences (DID) method, we analyze firms' environmental behavior under the ESG rating scheme and explore how social trust [2] and corruption moderate such behavior. We find that when faced with an imperfect ESG rating scheme, firms disproportionately pursue "symbolic" sustainability actions that significantly boost ESG scores at low cost and low risk, while avoiding "substantive" actions that require greater resources, involve higher risk, and deliver long-term returns. These tendencies are exacerbated by a dysfunctional social and political environment. A high-trust environment cultivates ethical norms that constrain managerial opportunism and diminish reliance on symbolic sustainability actions. In contrast, political corruption fosters rent-seeking, encourages symbolic environmental actions, and hinders sustainable corporate development.

This paper makes three main contributions to the literature. First, using a quasi-natural experiment, we find that greater investor and public attention to ESG ratings leads firms to prioritize symbolic over substantive sustainability actions, underscoring unintended distortions in ESG ratings and the need for reform to better reflect genuine sustainability goals. Second, applying impression management theory, we explain how dysfunctional social and political environments exacerbate firms' undesired behaviors. The conclusion reinforces the viewpoint of impression management theory, suggesting that in the absence of robust institutional frameworks, managers may resort to deliberately enhancing the firm's image to mask inappropriate behaviors. Third, in the early stages of ESG development in China, we document a clear disconnect between regulatory standards and corporate ESG practices, suggesting that policymakers should strengthen disclosure standards for sustainability reporting. Together, these findings provide important insights for regulating corporate sustainability behavior and advancing firms' genuine sustainable development.

2. Institutional background

Governments and the public's growing focus on sustainable development has driven a global transition towards green and sustainable development (Tian *et al.*, 2022). Investors now expect firms to demonstrate the long-term social and environmental impacts of their operations (Bromley and Powell, 2012). Managers increasingly view stronger environmental, social, and governance (ESG) performance as a means of enhancing reputation and competitiveness. Yet many firms pursue ESG primarily to secure legitimacy, leading to short-sighted "greenwashing" (Sauerwald and Su, 2019). Institutional theory holds that under legitimacy pressure, firms may engage in organizational decoupling—symbolically aligning with institutional norms while diverging substantively (Oliver, 1991). In this context, ESG

greenwashing constitutes a decoupling strategy, as ESG claims are detached from substantive ESG actions (Free *et al.*, 2024). This reflects a strategic response to the tension between external institutional demands and internal organizational practices.

Organization decoupling theory argues that when confronted with regulatory or policy demands, organizations may adopt symbolic compliance to meet public expectations, thereby separating formal requirements from actual practices (Meyer and Rowan, 1977). This explains the frequent gap between what firms say and what they actually do. Within the current ESG framework, firms may exaggerate their ESG disclosures to secure legitimacy, presenting information far beyond their actual performance—a strategic decoupling response to stakeholder pressure (Zhang, 2022). While this may sustain legitimacy in the short run, firms that rely on symbolic compliance face substantial “decoupling risks” as regulatory pressures tightens (Marquis and Qian, 2014). In the institutional literature, firms’ responses to environmental governance are commonly explained through three categories of institutional pressures: coercive, normative, and mimetic (Bag *et al.*, 2022). Coercive pressures stem from authoritative actors with the power to enforce compliance; normative pressures originate from professional norms and value systems; and mimetic pressures emerge when firms emulate the practices of leading peers (Tian *et al.*, 2025a). Collectively, these pressures shape corporate ESG practices and decoupling behaviors.

Studies show that investment and financing incentives foster synergies in corporate social responsibility: firms pursue ESG practices to limit ESG reputational risks, cultivate a positive image, and attract financing (Zhou *et al.*, 2024). Nonetheless, some studies have noted that ESG disclosure remains voluntary, with no uniform standard governing the content of reports. Without regulatory oversight, firms can selectively disclose or withhold operational information (Siew, 2015). Berg *et al.* (2022) further demonstrate that ESG ratings depend heavily on such disclosures, yet differ substantially across regions and rating agencies owing to variation in scope, measurement indicators, and weighting. This problem is particularly pronounced in emerging economies, where analysts’ optimistic bias often results in inflated ESG ratings and undermines their effectiveness in capturing ESG risks (Chen *et al.*, 2025). Therefore, the question remains: do firms disclosing ESG information genuinely undertake social responsibility, or are such practices primarily symbolic strategies aimed at obtaining financing?

ESG scores are tied primarily to corporate disclosure content rather than compliance history or actual environmental conduct. Some firms achieve high ratings by expanding disclosure without delivering substantive environmental or financial improvements (Raghunandan and Rajgopal, 2022). Others, despite lower overall emissions, may increase pollution to meet profit goals and rely on reputational strength to absorb the impact (Thomas *et al.*, 2022). Such behavior suggests that ESG ratings encourage symbolic rather than substantive ESG practices, reflecting greenwashing.

In 2015, China’s new *Environmental Protection Law* took effect and reinforced legislative sanctions on pollution and expanded public oversight. The revised law raised penalties for heavily polluting firms, increasing the costs of non-compliance and pushing firms toward green transition and improved ESG performance (Zhang and Liu, 2023). From an incentive design perspective, it mandated that firms establish sound ESG governance systems and enhance ESG disclosure, and provided rewards for strong ESG performers (Zhao *et al.*, 2024). At the social level, the law mandated disclosure by polluting firms and encouraged public participation, underscoring the critical supervisory role of media and the public (Wei *et al.*, 2025). Overall, the law shifted corporate ESG practice from passive compliance to proactive adjustment and promote the broader institutionalization of ESG principles in China’s market (He *et al.*, 2023). Since its implementation, ESG practices have drawn growing attention from government, business, and social media.

However, ESG ratings lack uniform standards, resulting in various rating schemes such as *HeXun* ESG, *CSI* ESG, *HuaZheng* ESG, *SynTao Green Finance*, and *LingRun* Global RKS. For example, in the early stages of ESG development in China, *HeXun* offered a quantitative

tool for assessing corporate ESG performance. *HeXun* ESG assesses firms' ESG practices based on annual reports and social responsibility reports from firms listed on the Shanghai and Shenzhen Stock Exchanges, covering shareholder, employee, supplier, customer, environmental, and social responsibilities (He *et al.*, 2022). Since the implementation of the "New Environmental Protection Law" in 2015, public attention to *HeXun* ESG ratings sharply increased, peaking in mid-2015 (as shown in Figure 1). Meanwhile, media reports on corporate ESG practices and related penalties surged in 2015 (Figure A1) [3].

Despite its broad acceptance among Chinese firms, investors, and media, the ESG rating scheme has significant flaws. In the *HeXun* model, innovation-related indicators are only assigned a total weight of 4–1% for product development expenditure, 1% for technological innovation ideas, and 2% for the number of innovation projects—while environmental awareness alone is weighted at 4%. Yet innovation typically involves higher costs, greater risks, and longer return cycle than environmental awareness. The *Huazheng model* adjusts indicator weights across industries, which can lead firms to channel resources into high-weighted areas while neglecting low-weight but high-risk ones. For instance, real estate firms may prioritize green building certification while overlooking investment in pollution abatement and greenhouse gas reduction. The *SynTao Green Finance* model, though comprising nearly 200 indicators, relies heavily on qualitative measures, creating opportunities for greenwashing. Firms can raise ESG scores through selective disclosure, ambiguous qualitative descriptions, and overstating positive outcomes. For example, a firm may claim to "actively engage in philanthropic activities" without specifying the projects, the scale of investment, or the beneficiary groups.

In practice, the absence of a strong judicial system weakens enforcement of China's environmental laws (Nguyen *et al.*, 2021), leading to poor implementation of environmental protection strategies (Du *et al.*, 2014). Flawed ESG rating schemes further enable opportunism, as firms use ESG reports to polish image and pursue managerial interests (Du, 2015). Through greenwashing, firms misrepresent their environmental principles and governance goals. Although ESG ratings are designed to encourage compliance and advance sustainability, many firms fail to meet their stated commitments (Huang *et al.*, 2020).

3. Hypothesis development

3.1 Corporate environmental decision-making under ESG rating schemes

ESG (Environmental, Social, and Governance) is often seen as a way for firms to prove legitimacy in the economic, social, and environmental realms, forming a social contract with their local community based on shared cultural, religious, and ethical values (Karwowski and Raulinajtys-Grzybek, 2021). In theory, firms engage in ESG for two main reasons: to further sustainable development goals and to serve managerial self-interest (Hao and He, 2022). Substantive engagement in ESG can raise productivity, improve competitiveness (Hur *et al.*, 2018), and, and, by enhancing environmental performance, build reputation and stakeholder



Figure 1. Web search index of HeXun. Source: Authors' own work

trust. In contrast, symbolic engagement reflects managers' pursuit of personal reputation, which may involve masking misconduct or leveraging ESG for career advancement (Li and Wu, 2020).

In China, ESG and socially responsible investing remain in their infancy, with development still at an early stage. Although regulatory authorities have issued guidance, several challenges remain, including insufficient disclosure, weak incentives for proactive reporting, and limited data availability (Luo *et al.*, 2023). Persistent problems in China's ESG rating schemes further undermine effectiveness. First, standards lack consistency across exchanges, industry groups, academic institutions, regulators, and rating agencies. Additionally, international indicators and weightings are often ill-suited to China's context. Second, the rating process lacks transparency and independence, generating inconsistent results across agencies and complicating evaluation by investors. Third, there's a lack of objective, structured, and quantifiable ESG data, leaving disclosures largely descriptive and difficult to measure.

The flaws in China's ESG rating schemes make it essential to investigate how firms respond to the rising investor and public scrutiny of ESG. Impression management theory suggests that managers can influence stakeholders' perceptions by cultivating a positive corporate image (Ashforth and Gibbs, 1990). Accordingly, some firms may emphasize favorable environmental portrayals while neglecting substantive initiatives (Wu *et al.*, 2022a). Prior studies further show that corporate social responsibility often serves as a tool to elevate stakeholder expectations and attract capital (Monfort *et al.*, 2021). In the evolving ESG landscape, firms may inflate ESG ratings by pledging ambitious environmental commitments. Such symbolic strategies require fewer resources than genuine action yet boost investor expectations through sustainability reports and future-oriented promises (Lee and Raschke, 2023).

Conversely, genuine sustainability efforts demand considerable efforts. Green innovation exemplifies such efforts, as it entails the development of eco-friendly products or production processes to reduce pollution and conserve resources (Yuan and Cao, 2022). It enables firms to meet environmental obligations and strengthen competitiveness, thereby achieving sustainable development (Hao and He, 2022). Yet, compared with other initiatives, green innovation is more costly, riskier, and slower to deliver returns. Consequently, under increased ESG scrutiny from investors and the public, firms may prioritize environmental activities that quickly boost ESG scores while avoiding investments in green innovation (Allcott and Greenstone, 2012). Therefore, we hypothesize:

- H1. Greater investor and public scrutiny of ESG ratings leads firms to prioritize symbolic sustainability activities that raise ESG scores over substantive sustainable actions.

3.2 Impact of social trust on corporate sustainability practices

China, as an emerging market, faces persistent gaps in environmental regulation. Social trust operates as an informal governance mechanism that significantly influences managerial choices and corporate strategies (Liu *et al.*, 2022a). Despite 3 decades of rapid economic expansion that have positioned China among the fastest-growing economies globally, social trust levels remain uneven across provinces and generally lower than those of Western developed nations. Surveys show that few view current social trust levels positively, with many perceiving a decline over time (Dong *et al.*, 2018). Promoting stronger social trust is therefore critical for the sustainable development of Chinese firms.

Research on social trust and ESG greenwashing reveals two contrasting views: trust exploitation (Gu *et al.*, 2022) and trust loyalty (Guan *et al.*, 2020). The trust exploitation perspective posits that social trust can create cognitive stereotypes, whereby firms in high-trust regions are presumed trustworthy. Some firms may take advantage of this presumption to avoid stakeholder scrutiny of environmental performance and engage in greenwashing through reputational signaling (Marquis *et al.*, 2016). By contrast, the trust loyalty perspective

contends that strong social trust fosters moral norms and heightens reputational risk (Haidar, 2025), encouraging firms to improve environmental performance and reducing incentives to greenwash (Shuang *et al.*, 2024). Thus, the impact of social trust on corporate ESG greenwashing remains insufficiently understood and requires deeper examination.

The paper argues that social trust is vital for well-functioning capital markets, as it reflects the level of cooperative orientation within society. On one hand, as an informal governance mechanism, social trust constrains corporate short-termism by fostering positive social norms that encourage ethical managerial behavior and by promoting information exchange among firms, employees, and stakeholders, thereby limiting opportunism (Ang *et al.*, 2015). Managers in high-trust regions exhibit greater honesty and reliability compared to their counterparts in low-trust regions (Choy *et al.*, 2023). Under ESG rating schemes, such managers are consequently less likely to rely on symbolic environmental pledges for impression management, instead favoring the fulfillment of commitments through substantive environmental action.

On the other hand, social trust mitigates information asymmetry, thereby increasing managers' sensitivity to reputation and the costs of opportunism (Dong *et al.*, 2018). It fosters behavioral expectations that crystallize into shared norms, shaping individual decision-making (Cialdini *et al.*, 1991). In regions with high social trust, reputational concerns weigh heavily on managerial careers (Liu *et al.*, 2022a). Firms pay a higher price for myopic actions, as reputational damage arises when managers prioritize personal gain over societal expectations. Under social scrutiny, managers are therefore less inclined to issue symbolic environmental pledges without substantive follow-through (Su and Song, 2022). Thus, in the context of rising ESG scrutiny from investors and the public, firms in high-trust regions are more likely to pursue genuine environmental strategies. Therefore, we hypothesize:

- H2. High social trust environments foster positive social norms that limit managerial opportunism, reducing firms' reliance on symbolic sustainability practices.

3.3 Impact of corruption on corporate sustainability practices

In China, business success often depends on social connections and political ties (Ru *et al.*, 2020), which some firms leverage to circumvent regulatory oversight. Decentralized environmental regulation grants local governments considerable discretion, creating space for corruption (Zhang, 2021). Coupled with rapid economic growth and weak institutional frameworks, this environment encourages bribery and rent-seeking (Xu *et al.*, 2019). Firms exploit these institutional gaps to gain political favors or policy advantages, frequently turning them into profits (Qi *et al.*, 2020). Data from the Central Commission for Discipline Inspection (CCDI) underscores the scale: in 2019, 20 centrally managed cadres, 59 cadres from central party and government organs, state-owned enterprises, and financial institutions, and 445 provincially managed cadres were investigated and disciplined.

Research also highlights the role of corruption in ESG greenwashing, as firms may use bribery to mask environmental misconduct. They may bribe government officials or media to suppress negative reports or fabricate positive images, misleading the public (Tella and Franceschelli, 2011). They may also bribe regulators to reduce enforcement intensity or evade penalties (Zhang, 2021). In addition, bribery may be used to manipulate product evaluations and certifications, weakening public oversight of greenwashing (Zhao *et al.*, 2023). Such practices erode the independence of governments, regulators, and media, making environmental violations harder to detect and reinforcing firms' incentives to greenwash (Wang *et al.*, 2025).

Corruption undermines China's environmental regulatory framework and jeopardizes economic sustainability. Polluting firms face high compliance costs to meet emission standards (Zhou *et al.*, 2022). To maximize profits, these firms often reduce compliance costs in two ways. First, some invest in green innovation to lower emission control costs, despite the associated high risks and costs. Others, exploiting the significant influence local governments

hold within China’s centralized political system (Hope *et al.*, 2020), resort to bribery to weaken enforcement (Wu *et al.*, 2022b). Such practices reduce production efficiency, incentivizes regulatory evasion, and ultimately diminishing long-term green investments (Bahoo *et al.*, 2020).

The “sanding-the-wheel” hypothesis in rent-seeking theory define corruption as the diversion of resources to secure privileges through political processes, resulting in public harm that outweighs the benefits accrued by the corrupt party (Rodríguez-Pose and Zhang, 2020; Paunov *et al.*, 2016). In China’s relatively immature institutional context, corruption erodes the effectiveness of central environmental regulations (Arminen and Menegaki, 2019). Firms engaging in corruption exacerbates environmental degradation by bribing officials as a means to evade environmental compliance, putting compliant firms at a disadvantage (Zhou *et al.*, 2022). Rent-seeking further redirects resources away from green innovation and governance, weakening managerial incentives for substantive environmental action. Consequently, in regions with high corruption, firms are more likely to engage in symbolic environmental actions—such as bribery and empty promises—to address investor and public concerns about environmental performance. Hence, we hypothesize:

- H3. Corruption intensifies managerial opportunistic motives, making firms more likely to engage in “tricky” sustainability activities when facing ESG ratings.

4. Research design

4.1 Data

This study uses data from Chinese firms listed on the Shanghai and Shenzhen Stock Exchanges from 2011 to 2019. Firms facing delisting (*ST stock) or under abnormal conditions (ST stock) are excluded. Financial and environmental governance data are sourced from the China Stock Market and Accounting Research (CSMAR) database, while additional information is gathered from the *China City Commercial Credit Environment Index White Paper* and China’s Corruption Investigations Dataset. Firms’ annual reports and news articles are reviewed to fill in missing data. The final dataset comprises an unbalanced panel of 9,306 firm-year observations across 1,188 listed firms.

4.2 Empirical design

To test our hypotheses, we use the January 2015 implementation of the new *Environmental Protection Law* as an exogenous shock to corporate sustainability practices and apply a Difference-in-Differences (DID) approach. To control for potential confounding factors such as time, region, and firm-level characteristics, we construct a two-way fixed-effects DID model to capture the Law’s effect on listed firms’ environmental strategies. The model is specified as follows:

$$Y_{it} = \beta_0 + \beta_1 Treat_i \times Post_t + \beta_2 Treat_i + \beta_3 Post_t + \gamma X_{it} + Firm_c + Year_t + \varepsilon_{it}$$

In the quasi-natural experiment, we use the interaction term $Treat_i \times Post_t$ (DID) as the explanatory variable to capture the impact of the implementation of the new *Environmental Protection Law* on corporate sustainability practices. Here, $Treat_i$ denotes the treatment effect. Compared with other industries, given that the law explicitly tightened environmental requirements and penalties for heavily polluting firms, and amid increasing public scrutiny of ESG performance, these firms are expected to pursue environmental actions more actively. Thus, listed firm in heavily polluting industries are classified as the treatment group (coded 1), while firms in other industries form the control group (coded 0). $Post_t$ captures the policy shock, taking a value of 1 for the year 2015 and onwards (when the *New Environmental Protection Law* took effect) and 0 otherwise. Y_{it} represents the key dependent variables (*Commit* and *Inno*), while X_{it} denote the vector of control variables.

4.3 Key variables measurement

Inno. We consider green innovation (**Inno**) as one of the key dependent variables. Green innovation, essential for sustainability (Zhang *et al.*, 2020), reduces pollution through new processes, methods, products, and services (Tian *et al.*, 2025b). Unlike conventional innovation, green innovation generates not only knowledge spillovers but also positive environmental spillovers such as resource conservation, carbon emission reduction, and greater energy efficiency (Chen *et al.*, 2021). Prior studies have employed input- and output-based indicators to capture green innovation, such as green R&D, green patents, and total factor productivity. Among these, patents are regarded as a more accurate measure of technological innovation. The World Intellectual Property Organization (WIPO) and the Organization for Economic Co-operation and Development (OECD) also define green innovation using patented technologies. Therefore, following Wu *et al.* (2022b), we measure green innovation by the total number of green patent applications filed by Chinese listed firms.

Commit. Environmental commitments and disclosure transparency are essential for evaluating ESG performance and signals a firm's legitimacy in environmental practices (Karwowski and Raulinajtyś-Grzybek, 2021). Environmental commitments act as positive signals of a firm's environmental behavior, reducing information gaps between investors, stakeholders, and the firm, and encouraging firms to prioritize environmental responsibility (Bolton *et al.*, 2021). Hence, we consider green commitments as a key dependent variable. These commitments are evaluated based on the dimensions such as air emissions reduction, water emissions reduction, industrial dust emissions reduction, and the management of industrial solid waste, as disclosed in annual or CSR reports. Our analysis shows that the most commonly disclosed items include the firm's environmental protection policies, annual environmental goals and outcomes, and the construction and operation of environmental facilities, making up over 70% of disclosures. In contrast, information on annual resource consumption, types of pollutant emitted, their quantity, concentration, and discharge direction are the least disclosed, appearing in fewer than 50% of disclosures. This discrepancy reflects firms' preference for symbolic commitments over information that may damage their reputation. Following Wiseman (1982), we code disclosures as follows: 0 if no air emission commitment is reported. Indicating no green commitment; 1 for vague goals lacking quantification, indicating symbolic commitment; and 2 for measurable emission reduction targets, indicating substantive commitment. The maximum score across dimensions is used as the proxy for Commit.

Following prior research, we control for financial and governance characteristics that may influence firms' environmental decisions and sustainability. At the firm level, we include controls for age, size, and performance. Firm age is measured by years since listing, as it correlates with environmental decision-making and sustainability. Firm size, proxied by total year-end assets, accounts for the greater reputational pressures faced by larger firms. Financial performance, measured by ROA, is also included, since prior studies suggest that firms with stronger profitability are more likely to engage in green innovation. Other factors such as leverage, Tb-Q, and ownership structure are also considered. With respect to governance variables, we control for board size, as larger boards enhance information flows and affect strategic outcomes. Ownership concentration, captured by the largest shareholder's stake, is included given its influence on decision-making. We also consider board independence, measured by the proportion of independent directors, due to their role in mitigating agency issues. Finally, we control for CEO duality, as combining CEO and chair roles may bias decisions toward personal rather than long-term corporate interests.

4.4 Descriptive statistics

Table 1 reports descriptive statistics and correlation matrix. *Commit* has a mean of 0.81 and a standard deviation of 0.8, reflecting relatively weak environmental commitments and limited disclosure transparency among Chinese listed firms. *Inno* has a mean of 0.41, with a range

Table 1. Descriptive statistics and correlations

	Mean	SD	Max	Min	1	2	3	4	5	6	7	8	9	10	11	12
1 <i>Inno</i>	0.41	0.92	6.87	0												
2 <i>Commit</i>	0.81	0.8	2	0	0.07***											
3 <i>Age</i>	17.74	5.72	44	7	−0.02*	0.05***										
4 <i>Size</i>	23.23	1.59	31.04	19.19	0.17***	0.20***	0.19***									
5 <i>ROA</i>	0.04	0.08	0.68	−1.75	0.01	−0.01	−0.07***	−0.09***								
6 <i>Leverage</i>	0.50	0.27	10.50	0.01	0.02**	0.02**	0.16***	0.45***	−0.42***							
7 <i>Tb_Q</i>	1.66	1.94	33.47	0.002	−0.03***	−0.12***	−0.1***	−0.36***	0.65***	−0.3***						
8 <i>SOE</i>	0.51	0.50	1	0	0.04***	0.14***	0.1***	0.28***	−0.09***	0.2***	−0.2***					
9 <i>Independ_ Director</i>	0.37	0.06	0.80	0.17	0.03	−0.01	−0.05***	0.06***	0.002	0.02*	0.04***	0.01				
10 <i>Largest_Share</i>	0.37	0.16	0.90	0.03	0.06***	0.09***	−0.14***	0.17***	0.05***	0.03***	−0.07***	0.29***	0.07***			
11 <i>Chairman_ Duality</i>	0.19	0.40	1	0	0.03***	−0.09***	−0.05***	−0.14***	0.05***	−0.08***	0.13***	−0.28***	0.09***	−0.1***		
12 <i>Board_Size</i>	9.19	2.09	21	3	0.06***	0.12***	0.07***	0.38***	−0.03***	0.17***	−0.16***	0.23***	−0.36***	0.001	−0.17***	

Note(s): This table summarizes descriptive statistics and correlations for the study's key variables. Columns (1)–(5) report the variable name, mean, standard deviation, maximum, and minimum, respectively; the remaining columns show correlation coefficients. The sample covers 9,306 firm-year observations of Chinese listed firms from 2011 to 2019

Source(s): Authors' own work

from 0 to 6.87 and a standard deviation of 0.92, suggesting heterogeneity across firms but relatively weak green innovation overall. Among the control variables, *Age* shows the highest variation, with a standard deviation of 5.72. The correlation analysis reveals statistically significant relationships between key dependent and control variables, suggesting that the ESG development of Chinese firms remains at a relatively early stage and is influenced by social conditions, firm maturity, and governance structure.

To clarify the distinction between symbolic and substantive behaviors across industries and capture heterogeneity in ESG responses, we analyze corporate sustainability practices of heavily polluting and non-polluting firms before and after the policy. As Table 2 reports, in the pre-policy period (2011–2014), heavily polluting firms ($N = 1,093$) had higher mean *Commit* (0.454 vs. 0.353) and *Inno* (0.901 vs. 0.523) values than non-polluting firms ($N = 2,453$), with both differences significant. In the post-policy period (2015–2019), heavily polluting firms ($N = 1,675$) recorded higher *Commit* (1.230 vs. 0.423) but lower *Inno* (0.709 vs. 0.426) than non-polluting firms ($N = 4,085$). The intergroup gap in *Commit* widened and was significant at the 1% level, while the gap in *Inno* narrowed, became insignificant, and even reversed. These results suggest that after the new *Environmental Protection Law*, heavily polluting firms increasingly relied on symbolic sustainability activities to boost ESG scores rather than engaging in substantive innovation, providing preliminary evidence for Hypothesis 1.

5. Empirical results

5.1 Baseline results

Table 3 displays the empirical results with robust standard errors. Model 1 and Model 2 show that since the enactment of the new *Environmental Protection Law*, rising investor and the public scrutiny of ESG ratings significantly increased firms' environmental commitments ($\beta = 0.082, p < 0.05$) but reduced green innovation investment ($\beta = -0.076, p < 0.05$). This supports Hypothesis 1, suggesting that under imperfect ESG ratings, firms are more likely to engage in symbolic sustainability activities to improve ESG scores rather than making substantive changes when investors and the public focus more on ESG (Environmental, Social, and Governance) ratings, reflecting opportunism. To disentangle the nature of green innovation, patents are classified into green invention patents, which reflect breakthrough contributions involving significant technological advancements, and green utility model patents, which reflect incremental adjustments to existing technologies. Following Sun *et al.* (2021), we classify green innovation (*Inno*) into breakthrough green

Table 2. Descriptive statistics of corporate sustainability practices by industry

	2011–2014 (1)	(2)	(3)	2015–2019 (4)	(5)	(6)
	Heavily polluting industries	Non- heavily polluting industries	<i>t</i> -test	Heavily polluting industries	Non- heavily polluting industries	<i>t</i> -test
<i>Commit</i>	0.454	0.353	3.195***	1.230	0.709	22.758***
<i>Inno</i>	0.901	0.523	14.270***	0.423	0.426	−0.078
<i>N</i>	1,093	2,453		1,675	4,085	

Note(s): This table shows descriptive statistics for corporate sustainability practices in heavily polluting and non-heavily polluting industries before (2011–2014) and after (2015–2019) the policy. Columns (1) and (4) report the means of *Commit* and *Inno* and sample sizes for heavily polluting industries across the two periods; Columns (2) and (5) report the corresponding statistics for non-heavily polluting industries; and Columns (3) and (6) reports the results of inter-group difference tests. *** indicates significance at the 1% level (two-tailed)

Source(s): Authors' own work

Table 3. Baseline results

Variables	Model 1 Commit	Model 2 Inno
<i>DID</i>	0.082** (0.038)	−0.076** (0.031)
<i>Age</i>	0.042*** (0.006)	−0.024*** (0.004)
<i>Size</i>	0.038 (0.024)	0.056*** (0.020)
<i>ROA</i>	−0.024 (0.125)	0.111 (0.102)
<i>Leverage</i>	−0.099 (0.074)	−0.085 (0.068)
<i>Tb_Q</i>	−0.001 (0.006)	−0.012*** (0.004)
<i>SOE</i>	−0.037 (0.059)	0.102* (0.054)
<i>Independ_Director</i>	−0.001 (0.241)	−0.126 (0.233)
<i>Largest_Share</i>	0.024 (0.160)	−0.127 (0.114)
<i>Chairman_Duality</i>	−0.054* (0.030)	0.012 (0.025)
<i>Board_Size</i>	−0.004 (0.011)	0.009 (0.010)
<i>Constant</i>	−0.678 (0.540)	−0.633 (0.428)
<i>Observations</i>	8,822	9,218
<i>Firm FE</i>	Yes	Yes
<i>Year FE</i>	Yes	Yes
<i>R-squared</i>	0.071	0.074

Note(s): This table presents DID regression results of the effects of the new Environmental Protection Law on corporate sustainability practices. Column (1) presents the effect on corporate environmental commitments; Column (2) presents the effect on corporate green innovation. Standard errors are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively

Source(s): Authors' own work

innovation (*Green_Invention*) and incremental green innovation (*Green_Practical*) to assess post-policy effects. Regression results show that the implementation of the new *Environmental Protection Law* significantly reduced investment in breakthrough green innovation ($\beta = -0.055, p < 0.1$), while its impact on incremental green innovation investment was insignificant. This suggests firms prioritize symbolic environmental actions to appease stakeholders and regulators expectations (Delmas and Burbano, 2011), while scaling back substantive environmental efforts.

5.2 Moderating effect

We also examine the moderating effects of social trust and corruption. Social trust, as an informal institution, prompts managers to prioritize corporate reputation, reducing managerial opportunism and myopic behavior (Liu et al., 2022a). Using the China City Commercial Credit Environment Index, we measure social trust in the regions where firms are located to examine its impact on firms' environmental governance decisions. Table 4 reports the results for the moderating effect of social trust. Results from Model 1–4 support Hypothesis 2: in regions with low social trust, firms focus more on short-term gains, increasing environmental commitments ($\beta = 0.099, p < 0.05$) but reducing green innovation investment ($\beta = -0.083$,

Table 4. Moderating effect

Variables	Social trust				Corruption			
	Model 1 High Commit	Model 2 Low Commit	Model 3 High Inno	Model 4 Low Inno	Model 5 High Commit	Model 6 Low Commit	Model 7 High Inno	Model 8 Low Inno
<i>DID</i>	0.043 (0.063)	0.099** (0.049)	−0.100 (0.062)	−0.083** (0.042)	0.107** (0.054)	0.023 (0.054)	−2.619** (1.231)	1.025 (0.970)
<i>Age</i>	0.052*** (0.008)	0.039*** (0.008)	−0.020*** (0.006)	−0.026*** (0.006)	0.007 (0.016)	0.037*** (0.011)	0.908** (0.400)	0.384*** (0.112)
<i>Size</i>	0.024 (0.032)	0.028 (0.036)	0.035 (0.028)	0.067** (0.028)	0.021 (0.046)	−0.013 (0.034)	1.224** (0.504)	−0.087 (0.206)
<i>ROA</i>	0.314** (0.156)	−0.268 (0.167)	0.130 (0.144)	0.195 (0.145)	−0.279 (0.376)	−0.009 (0.215)	−0.156 (4.935)	2.171 (1.611)
<i>Leverage</i>	−0.003 (0.080)	−0.056 (0.137)	−0.047 (0.091)	−0.194** (0.087)	−0.196 (0.165)	−0.267** (0.134)	−10.213*** (3.444)	−0.890 (1.332)
<i>Tb_Q</i>	−0.000 (0.006)	−0.003 (0.010)	−0.011* (0.006)	−0.016** (0.007)	−0.009 (0.011)	−0.024*** (0.007)	−0.084 (0.134)	−0.164*** (0.045)
<i>SOE</i>	−0.125 (0.095)	0.071 (0.075)	0.167** (0.075)	0.073 (0.079)	−0.037 (0.093)	0.039 (0.136)	2.173 (1.484)	0.664 (0.452)
<i>Independ_Director</i>	0.381 (0.312)	−0.281 (0.399)	−0.017 (0.378)	−0.299 (0.303)	0.132 (0.550)	−0.229 (0.389)	−6.695 (7.183)	−1.726 (4.017)
<i>Largest_Share</i>	0.132 (0.218)	0.036 (0.241)	0.078 (0.210)	−0.188 (0.139)	−0.296 (0.317)	0.079 (0.267)	7.781 (7.691)	0.893 (1.326)
<i>Chairman_Duality</i>	−0.092* (0.048)	−0.009 (0.037)	0.026 (0.037)	−0.012 (0.034)	−0.054 (0.058)	−0.029 (0.049)	1.554 (1.742)	−0.044 (0.262)
<i>Board_Size</i>	0.012 (0.015)	−0.014 (0.016)	0.013 (0.017)	0.007 (0.011)	0.025 (0.021)	0.003 (0.016)	0.097 (0.488)	−0.320 (0.393)
<i>Constant</i>	−0.978 (0.711)	−0.181 (0.814)	−0.434 (0.652)	−0.647 (0.591)	0.188 (1.023)	0.579 (0.777)	−36.029** (15.375)	2.600 (7.399)
<i>Observations</i>	4,300	4,406	4,490	4,609	2,626	2,727	2,759	2,991
<i>Firm FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>R-squared</i>	0.069	0.076	0.082	0.070	0.024	0.020	0.033	0.027

Note(s): This table presents the analysis of moderating effects. Columns (1)–(4) show the moderating role of social trust in the baseline regressions, whereas Columns (5)–(8) report the moderating role of regional corruption. Standard errors are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively

Source(s): Authors' own work

$p < 0.05$). Conversely, this trend is less pronounced in regions with higher social trust, suggesting that low social trust exacerbates myopic behavior among managers, hindering corporate sustainable development. The results imply that, amid greater public attention to ESG, social trust incentivizes firms to pursue substantive green strategies instead of resorting to greenwashing.

Political bribery a common non-market strategy in emerging markets (Ufere *et al.*, 2020). Firms engage in political bribery to gain privileged information or policy benefits, resulting in lucrative business opportunities (Zhang, 2021). To examine its impact on corporate environmental governance decisions, we measure regional corruption by counting the number of corrupt officials in each province of China. The moderating effect of corruption is shown in Table 4. Results from Model 5 to 8 reveal that in high-corruption regions, firms focus on symbolic environmental activities, increasing environmental commitments ($\beta = 0.107$, $p < 0.05$) but reducing investment in green innovation ($\beta = -2.619$, $p < 0.05$). This highlights how corruption drives managerial opportunism and myopic decisions, such as engaging in symbolic sustainability activities in response to ESG ratings, confirming Hypothesis 3.

6. Robustness tests

6.1 Parallel trend analysis

We apply a DID model to estimate the impact of the new *Environmental Protection Law* on corporate sustainability practices. Figure 2 visualizes the parallel trends and dynamic effects. The left panel shows that before the policy, all coefficients are insignificant, indicating that treatment and control firms followed similar trajectories in green commitments. After the policy, coefficients become significantly positive, implying that treatment firms improved commitments more than controls, with effects intensifying over time. The right panel shows parallel pre-policy trends in green innovation, while post-policy coefficients are significantly negative, suggesting that treatment firms reduced green innovation more sharply than controls.

6.2 Placebo test

To mitigate potential bias from unobserved omitted factors, we conduct a placebo test following Tian *et al.* (2024). We reset the policy year to 2013, reconstruct the *Post* variable, and re-estimate the impact of the new *Environmental Protection Law* on corporate sustainability practices. As reported in Table 5, columns (1) and (2), the DID coefficients are insignificant, confirming the robustness of our baseline results.

We also perform a permutation-based placebo test following Liu *et al.* (2022c). Firms are randomly assigned to treatment, policy years are randomly generated, and placebo regressions are run using the baseline model. The procedure is repeated 1,000 times, and the distribution of DID coefficients is plotted. As shown in Figure 3, the placebo estimates cluster around zero in

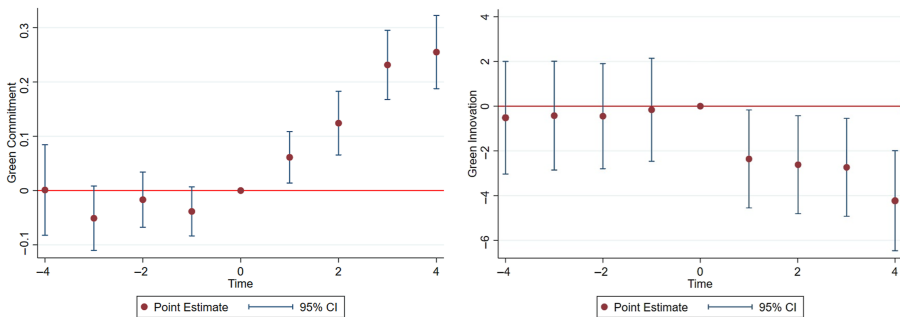


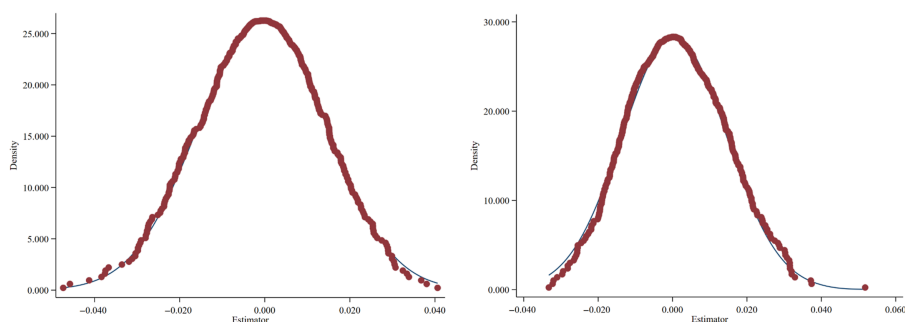
Figure 2. Parallel trend analysis. Source: Authors' own work

Table 5. Placebo test

Variables	Model 1 Commit	Model 2 Inno
<i>DID</i>	0.048 (0.043)	−0.042 (0.035)
<i>Age</i>	0.044*** (0.006)	−0.026*** (0.004)
<i>Size</i>	0.033 (0.024)	0.061*** (0.020)
<i>ROA</i>	−0.012 (0.125)	0.101 (0.102)
<i>Leverage</i>	−0.097 (0.074)	−0.086 (0.068)
<i>Tb_Q</i>	−0.001 (0.006)	−0.012*** (0.004)
<i>SOE</i>	−0.040 (0.060)	0.104* (0.054)
<i>Independ_Director</i>	−0.003 (0.242)	−0.126 (0.233)
<i>Largest_Share</i>	0.026 (0.160)	−0.129 (0.114)
<i>Chairman_Duality</i>	−0.052* (0.030)	0.010 (0.025)
<i>Board_Size</i>	−0.003 (0.011)	0.009 (0.010)
<i>Constant</i>	−0.597 (0.539)	−0.703 (0.429)
<i>Observations</i>	8,822	9,218
<i>Firm FE</i>	Yes	Yes
<i>Year FE</i>	Yes	Yes
<i>R-squared</i>	0.070	0.073

Note(s): This table presents the placebo test results. We advance the policy implementation year by two years (to 2013) and redefine the Post variable for the analysis. Standard errors are reported in parentheses; ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively

Source(s): Authors' own work

**Figure 3.** Placebo test. Source: Authors' own work

an inverted U-shape, far from our baseline coefficients of 0.082 (*Commit*) and −0.076 (*Inno*). This confirms that the baseline results are not attributable to random unobserved factors, reinforcing their robustness.

6.3 Alternative measures

To validate the robustness of our findings, we conduct four tests using alternative measures of the key variables (Table 6 Panel A). First, we replace the environmental commitment variable (*Commit*) with whether listed firms disclose pollutants in their operations (*Alter_Commit*) and run logistic regression, controlling for firm and year fixed effects. The results show that when investors and the public focus more on ESG (Environmental, Social, and Governance) ratings, listed firms are more inclined to disclose environmental information ($\beta = 0.454, p < 0.01$) to improve their ESG scores. Second, we substitute whether listed firms disclose their environmental concepts, policies, and green development models (*EP_concept*) for environmental commitment (*Commit*) and re-estimate the regression. The results show that, in response to ESG rating, listed firms are more likely to disclose their environmental concepts ($\beta = 0.306, p < 0.05$) to enhance their pro-environment image. To further assess substantive green behavior, we replace green innovation (*Inno*) with environmental investment (*Env-inv*). The results show that under the current ESG rating scheme, listed firms are more inclined to reduce environmental investment ($\beta = -0.355, p < 0.01$) to cut costs. We further substitute green innovation (*Inno*) with pollution emissions (*Pollution*) to evaluate whether the new *Environmental Protection Law* has curbed emissions. The results indicate that pollution among listed firms continues to rise, with no substantive reduction ($\beta = 0.014, p < 0.01$). These results reinforce the robustness of our baseline results.

6.4 Measurement of greenwashing

Additionally, we construct a combined measure of greenwashing using percentiles of the key variables, environmental commitment (*Commit*) and corporate green innovation (*Inno*), for regression analysis (Table 6 Panel B). In Model 1, we group *Commit* and *Inno* by year, calculate their percentiles, and subtract them to form *Rank1*. In Model 2, we convert the variables into binary ones based on the median and subtract them to form *Rank2*. In Models 3 and 4, we group the variables by industry, calculate their percentiles, and form *Rank3* and *Rank4*. The regression results in Table 6 show significant and positive coefficients for the DID variables, indicating that when the public focus more on ESG ratings, listed firms are more likely to polish their environmental image through greenwashing.

6.5 Time-varying fixed effects

Unobserved time-varying regional and industry factors may bias the estimates. Because industrial activity is geographically clustered, agglomeration effects may shape corporate sustainability practices. Geographic heterogeneity may also correlate with regulatory stringency; for example, local green subsidy policies or environmental enforcement regimes can affect firms' compliance costs and decisions on environmental violations or green R&D. To mitigate such concerns, and following Ellis *et al.* (2020), we add $Year \times Province$ and $Year \times Industry$ interaction fixed effects to the baseline model. The results are shown in Table 6, Panel C. Models 1–2 include firm and $Year \times Province$ fixed effects, while Models 3–4 include firm and $Year \times Industry$ fixed effects. The coefficients for *Commit* remain positive and significant, while those for *Inno* remain negative and significant. These results suggest that our findings are robust and not driven by time-varying regional or industry factors.

6.6 PSM-DID

To ensure sample consistency, this study uses propensity score matching (PSM) to match the samples. Following existing literature, the matching variables include *Age*, *Size*, *ROA*, *Leverage*, *Tb_Q*, *SOE*, *Independ_Director*, *Largest_Share*, *Chairman_Duality* and *Board_Size*. Using a nearest neighbor 1-to-1 matching method, we derive the matched sample. The balancing test of the matched sample shows that the standard deviations of all variables are below 10%, with insignificant *p*-values. This suggests no meaningful differences between the treated and control groups, confirming matching quality. Figure 4 displays the kernel density curves of propensity scores for the treatment and control groups.

Table 6. Robustness tests

Panel A: Alternative measures				
Variables	Model 1 Alter_ Commit	Model 2 EP_concept	Model 3 Env-inv	Model 4 Pollution
<i>DID</i>	0.454*** (0.171)	0.306** (0.155)	−0.355*** (0.112)	0.014*** (0.005)
<i>Age</i>	−0.151 (0.120)	−0.038 (0.182)	0.030 (0.038)	0.132*** (0.001)
<i>Size</i>	0.174 (0.159)	0.894*** (0.127)	0.766*** (0.214)	0.011*** (0.004)
<i>ROA</i>	0.481 (0.873)	−0.176 (0.660)	−0.268 (1.081)	−0.053** (0.026)
<i>Leverage</i>	0.600 (0.569)	−0.160 (0.431)	0.918 (0.748)	−0.006 (0.014)
<i>Tb_Q</i>	−0.153** (0.060)	0.067** (0.033)	0.111* (0.058)	0.006*** (0.001)
<i>SOE</i>	−0.072 (0.470)	0.101 (0.324)	0.397 (0.439)	−0.008 (0.010)
<i>Independ_Director</i>	0.940 (1.227)	−0.337 (1.050)	2.258** (1.036)	−0.001 (0.039)
<i>Largest_Share</i>	−0.394 (0.785)	1.323* (0.694)	0.465 (0.952)	−0.008 (0.022)
<i>Chairman_Duality</i>	−0.346** (0.165)	−0.180 (0.137)	0.181 (0.170)	0.008 (0.005)
<i>Board_Size</i>	0.059 (0.050)	0.059 (0.044)	0.124** (0.056)	−0.002 (0.001)
<i>Constant</i>	−4.178 (3.838)	−24.266*** (3.430)	−5.520 (4.833)	11.659*** (0.080)
<i>Observations</i>	4,600	6,433	1,124	9,181
<i>Firm FE</i>	Yes	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes	Yes
<i>R-squared</i>			0.079	0.154

Note(s): This table presents robustness checks for the effect of the new Environmental Protection Law on corporate sustainability practices. Columns (1) and (2) report regressions with alternative measures of symbolic actions, and Columns (3) and (4) report regressions with alternative measures of substantive actions. Standard errors are reported in parentheses; ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively

Source(s): Authors' own work

Panel B: Measurement of greenwashing				
Variables	Model 1 Rank1	Model 2 Rank2	Model 3 Rank3	Model 4 Rank4
<i>DID</i>	0.047** (0.020)	0.129*** (0.030)	0.049** (0.019)	0.101*** (0.029)
<i>Age</i>	0.020*** (0.003)	0.056*** (0.005)	0.166*** (0.012)	0.190*** (0.017)
<i>Size</i>	−0.013 (0.012)	−0.045*** (0.016)	−0.002 (0.012)	−0.005 (0.017)
<i>ROA</i>	−0.127* (0.067)	−0.037 (0.094)	−0.104 (0.065)	−0.133 (0.097)
<i>Leverage</i>	−0.028 (0.042)	−0.002 (0.056)	−0.031 (0.042)	−0.042 (0.059)
<i>Tb_Q</i>	0.002 (0.002)	−0.002 (0.004)	0.002 (0.002)	0.004 (0.004)
<i>SOE</i>	−0.057 (0.037)	−0.074 (0.050)	−0.065* (0.038)	−0.063 (0.050)
<i>Independ_Director</i>	0.049 (0.135)	0.025 (0.183)	0.016 (0.133)	−0.064 (0.184)

(continued)

Table 6. Continued

Panel B: Measurement of greenwashing				
Variables	Model 1 Rank1	Model 2 Rank2	Model 3 Rank3	Model 4 Rank4
<i>Largest_Share</i>	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
<i>Chairman_Duality</i>	−0.028* (0.016)	−0.034 (0.022)	−0.022 (0.016)	−0.017 (0.023)
<i>Board_Size</i>	−0.004 (0.006)	0.002 (0.008)	−0.005 (0.006)	−0.004 (0.008)
<i>Constant</i>	0.138 (0.278)	0.109 (0.371)	−2.045*** (0.307)	−2.364*** (0.419)
<i>Observations</i>	8,762	8,762	8,762	8,762
<i>Firm FE</i>	Yes	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes	Yes
<i>R-squared</i>	0.011	0.045	0.068	0.062

Note(s): Columns (1) and (2) report regressions based on data grouped by year, with percentiles calculated for Commit and Inno. Columns (3) and (4) report regressions using data grouped by industry, with percentiles calculated and year fixed effects controlled for in the analysis. Standard errors are reported in parentheses; ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively

Source(s): Authors' own work

Panel C: Time-varying fixed effects				
Variables	Model 1 Commit	Model 2 Inno	Model 3 Commit	Model 4 Inno
<i>DID</i>	0.048* (0.028)	−0.073** (0.033)	0.368*** (0.141)	−0.161** (0.071)
<i>Age</i>	0.047 (0.188)	−0.044 (0.071)	0.115 (0.187)	−0.049 (0.036)
<i>Size</i>	0.036* (0.019)	0.061*** (0.020)	0.031 (0.026)	0.053** (0.023)
<i>ROA</i>	−0.009 (0.126)	0.109 (0.104)	−0.017 (0.128)	0.079 (0.122)
<i>Leverage</i>	−0.078 (0.068)	−0.079 (0.071)	−0.120 (0.075)	−0.069 (0.090)
<i>Tb_Q</i>	−0.002 (0.006)	−0.010** (0.005)	−0.003 (0.007)	−0.013** (0.005)
<i>SOE</i>	−0.060 (0.053)	0.111** (0.056)	−0.036 (0.069)	0.063 (0.053)
<i>Independ_Director</i>	0.046 (0.192)	−0.096 (0.239)	−0.078 (0.239)	0.053 (0.250)
<i>Largest_Share</i>	−0.004 (0.114)	−0.096 (0.119)	0.034 (0.159)	−0.111 (0.119)
<i>Chairman_Duality</i>	−0.064*** (0.024)	0.015 (0.026)	−0.038 (0.030)	0.006 (0.025)
<i>Board_Size</i>	−0.005 (0.008)	0.010 (0.010)	−0.006 (0.010)	0.013 (0.010)
<i>Constant</i>	−0.726 (3.906)	−0.373 (1.530)	−2.086 (4.054)	0.081 (0.853)
<i>Observations</i>	8,822	9,218	8,822	9,218
<i>Firm FE</i>	Yes	Yes	Yes	Yes
<i>Year-Province FE</i>	Yes	Yes	No	No
<i>Year-Industry FE</i>	No	No	Yes	Yes
<i>R-squared</i>	0.106	0.104	0.155	0.190

Note(s): Columns (1) and (2) present regression results with firm fixed effects and year–province interaction fixed effects, while Columns (3) and (4) present results with firm fixed effects and year–industry interaction fixed effects. Standard errors are reported in parentheses; ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively

Source(s): Authors' own work

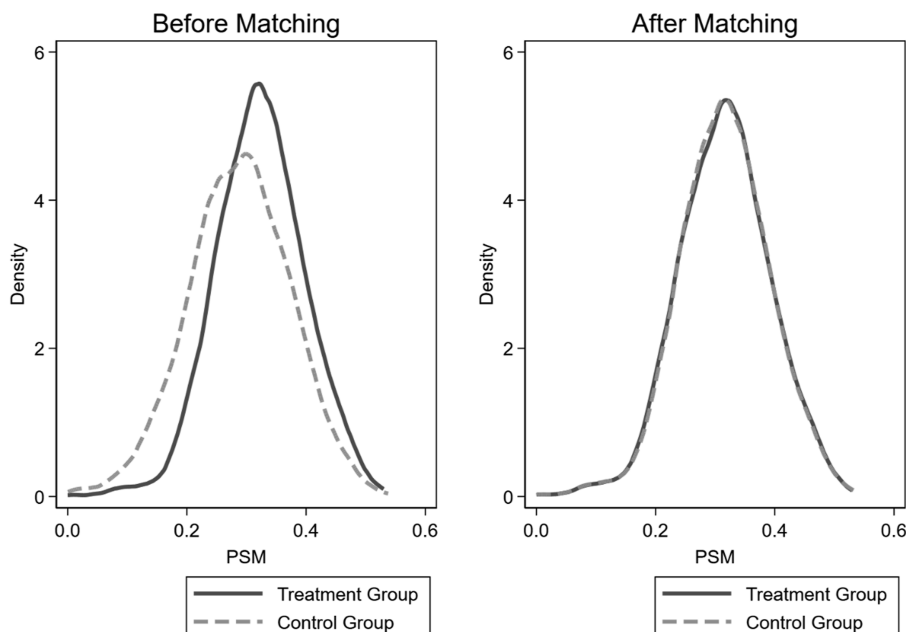


Figure 4. Kernel density distribution of propensity scores for the treatment and control groups before and after matching. Source: Authors' own work

Figure 4 shows that before PSM, the control group shows a left-skewed distribution and greater dispersion than the treatment group, with a significant difference in probability density distribution. After nearest-neighbor 1-to-1 matching, the probability density distributions of the two groups align, satisfying the DID assumption. Table 7 presents regression results from the matched sample. Using the matched environmental commitment (*_Commit*) as the dependent variable, the DID coefficient in Column (1) is ($\beta = 0.083, p < 0.05$). In Column (2), with matched green innovation (*_Inno*) as the dependent variable, the DID coefficient is ($\beta = -0.075, p < 0.05$). This suggests that greater investor and public attention to ESG prompts firms to engage in greenwashing via ESG rating schemes, validating the robustness of the baseline results.

6.7 Difference-in-difference-in-differences (DDD)

A remaining concern is whether industry-specific weighting schemes in ESG ratings bias our results. For example, the *HeXun* system assigns environmental responsibility weights of 30% to manufacturing, 10% to services, and 20% to other industries. To mitigate this concern, we construct a difference-in-difference-in-differences (DDD) model using an industry-weight variable (*Ind-weights*), set at 0.3 for manufacturing firms, 0.1 for service firms, and 0.2 for others. This specification controls for industry weighting and isolates the net effect of the new *Environmental Protection Law* on corporate sustainability practices. Table 8 shows that, even after accounting for these factors, our baseline results remain highly robust.

7. Discussion and conclusion

To investigate whether ESG (Environmental, Social, and Governance) ratings impact sustainability in Chinese listed firms, this study uses the 2015 enactment of the *New Environmental Protection Law* as an exogenous shock to investor and public attention to ESG.

Table 7. PSM-DID

Variables	Model 1 _Commit	Model 2 _Inno
<i>DID</i>	0.083** (0.038)	−0.075** (0.031)
<i>Age</i>	0.043*** (0.006)	−0.024*** (0.004)
<i>Size</i>	0.037 (0.024)	0.057*** (0.020)
<i>ROA</i>	−0.017 (0.125)	0.117 (0.103)
<i>Leverage</i>	−0.100 (0.074)	−0.086 (0.069)
<i>Tb_Q</i>	−0.003 (0.006)	−0.013*** (0.005)
<i>SOE</i>	−0.037 (0.060)	0.101* (0.054)
<i>Independ_Director</i>	−0.001 (0.241)	−0.124 (0.233)
<i>Largest_Share</i>	0.026 (0.161)	−0.123 (0.114)
<i>Chairman_Duality</i>	−0.055* (0.030)	0.011 (0.025)
<i>Board_Size</i>	−0.003 (0.011)	0.009 (0.010)
<i>Constant</i>	−0.673 (0.540)	−0.630 (0.428)
<i>Observations</i>	8,818	9,214
<i>Year FE</i>	Yes	Yes
<i>Firm FE</i>	Yes	Yes
<i>R-squared</i>	0.071	0.074

Note(s): This table presents results of the PSM-DID analysis. Standard errors are reported in parentheses; ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively

Source(s): Authors' own work

It examines corporate environmental behavior under ESG rating schemes. We argue that ESG ratings are used for firms to build a positive image and attract investment, drawing from concepts like impression management and opportunism theory. Motivated by impression management, firms highlight positive information while concealing negatives to capture investor and public attention (Liu *et al.*, 2022b). For example, some energy firms may prioritize financial optics over true sustainability investment, using strategies like earnings management to embellish financial prospects and downplay environmental issues (Cao *et al.*, 2023). This has raised public doubt about the authenticity of corporate environmental commitments and green initiatives (Winn and Angell, 2000).

China, the world's second-largest economy and largest emitter of pollutant, struggles with its incomplete environmental regulations and weak enforcement of environmental laws. While many firms have demonstrated a positive environmental image, some still engage in excessive greenwashing. Despite the introduction of environmental laws, their enforcement is undermined by a lack of effective judicial oversight (Du *et al.*, 2014). Moreover, ethical standards among Chinese listed firms are remain under development, limiting their proactive engagement in environmentally friendly practices (Du, 2015). This context provides an opportunity to investigate the impact of ESG rating schemes on the environmental behavior of Chinese firms.

Table 8. Difference-in-Difference-in-differences (DDD)

Variables	Model 1 Commit	Model 2 Inno
<i>DDD</i>	0.616*** (0.119)	−0.284*** (0.104)
<i>Ind-weights</i>	2.575*** (0.186)	2.254*** (0.237)
<i>Age</i>	0.001 (0.003)	−0.016*** (0.004)
<i>Size</i>	0.104*** (0.012)	0.114*** (0.018)
<i>ROA</i>	−0.140 (0.121)	0.080 (0.106)
<i>Leverage</i>	−0.196*** (0.065)	−0.099 (0.068)
<i>Tb_Q</i>	−0.006 (0.005)	−0.007* (0.004)
<i>SOE</i>	0.131*** (0.032)	0.082** (0.036)
<i>Independ_Director</i>	0.109 (0.199)	−0.065 (0.211)
<i>Largest_Share</i>	0.133 (0.096)	−0.138 (0.097)
<i>Chairman_Duality</i>	−0.078*** (0.026)	0.022 (0.025)
<i>Board_Size</i>	0.009 (0.008)	0.010 (0.009)
<i>Constant</i>	−0.597 (0.539)	−0.703 (0.429)
<i>Observations</i>	8,822	9,218
<i>Firm FE</i>	Yes	Yes
<i>Year FE</i>	Yes	Yes
<i>R-squared</i>	0.184	0.132

Note(s): This table presents the results of the DDD analysis. Ind-weights denote the environmental weights in the ESG rating framework, set at 0.3 for manufacturing, 0.1 for services, and 0.2 for all other industries. Standard errors are reported in parentheses; ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively

Source(s): Authors' own work

In China's emerging market, we conduct a quasi-natural experiment to investigate how firms shape their sustainable strategies in response to growing investor interest in ESG. Our findings show, first, that when faced with ESG ratings, firms prioritize “tricky” sustainability actions for higher ESG scores, with lower costs and risks, rather than investing in more substantive environmental governance with higher costs, risks, and longer payback periods. Second, high social trust promotes better ethical standards, reducing “tricky” sustainability activities. Third, rampant corruption fuels managerial rent-seeking behavior and symbolic environmental actions, hindering sustainable development.

Our research contributes to existing literature in two key ways. First, we find that in China's early ESG development stage, managers see ESG ratings as a tool for shaping corporate image. Most previous studies have focused on how managers used financial management or internal governance adjustment to enhance firm image. For example, to manage stakeholder impressions, firms often use channels like advertising and media to demonstrate compliance (Du, 2015). However, as sustainable development gains traction, ESG behavior has become a way to strengthen stakeholder relationships (He *et al.*, 2022). Our study expands on impression

management theory by examining its application in ESG practices, providing fresh insights into how ESG ratings impact corporate environmental decisions.

Second, this study explores undesired sustainability behaviors among firms with high ESG ratings through the lens of stakeholder salience theory. We show that negative social and political environments heighten managerial incentives for such behavior. We apply social trust theory to explain how high social trust environments promotes ethical business norms and reduce managerial myopia. Moreover, within China's relationship-based society, we explore the strategies firms use to navigate environmental regulations. Drawing on rent-seeking theory, we demonstrate how corruption fuels managerial myopia.

Finally, our study has significant practical implications. We find that the imperfect ESG rating scheme, weak regulatory enforcement, and the lack of ethical norms in firms drive opportunistic behavior that inflates ESG scores. Based on these findings, we propose several policy recommendations. First, environmental regulation should avoid a uniform "one-size-fits-all" approach. Governments should design targeted and supportive legal frameworks by developing robust evaluation and assessment systems for environmental governance, thereby providing a legal foundation for corporate ESG practices. Second, governments should mandate corporate environmental information disclosure and establish a unified ESG evaluation system. To complement this, third-party rating agencies should disclose their indicators and methodologies, as greater transparency would enable independent scrutiny and reduce bias in ESG ratings. Third, the regulatory system should be strengthened by introducing public accountability mechanisms and linking local firms' unfair competition practices to the performance evaluation of local governments, ensuring consistent and fair enforcement of environmental laws. Finally, governments should develop an environmental credit system, incorporating greenwashing into corporate credit ratings to increase the costs and exposure of such misconduct. Tackling greenwashing at its root is crucial for effective environmental regulation, improved corporate credibility, and sustainable development. This research deepens the understanding of the factors influencing corporate environmental behavior under ESG rating schemes and offers essential insights for improving these schemes, regulating the environmental practices of listed firms, and promoting economic sustainability.

8. Limitations and avenues for future research

This study also has several limitations that point to avenues for future research. Our analysis focuses on China, where environmental regulation is relatively recent and where the definition, identification, and sanctioning of greenwashing remain ambiguous. This ambiguity enables firms to exploit loopholes, making it harder to identify greenwashing (Hu *et al.*, 2023). Currently, there is no consensus in academia and society on how to define and measure corporate greenwashing. Existing approaches rely either on media or stakeholder assessments (Capelle-Blancard and Petit, 2019), or on environmental disclosed-based proxies, which typically assume that overstating commitments relative to actions indicates greenwashing. Yet, undisclosed activities cannot be observed, so measurement inevitably relies on reported information (Wang and Liu, 2024). Our study uses the quality of environmental disclosed and levels of green innovation as proxies, but the accurate identification of greenwashing remains challenging. As regulatory oversight and sanctioning systems evolve, future research can exploit case-level enforcement data to improve measurement precision. For instance, Huahong Steel (000,932) claimed to pursue "green development" in its corporate narrative in 2023, yet several subsidiaries lacked pollution control facilities, were penalized for severe pollution [4]. Future research could use data from such punishment cases to improve the accuracy of identifying greenwashing.

In addition, beyond social factors such as social trust and corruption, industry regulation and uneven regional enforcement also influence greenwashing, producing variation in its manifestation across industries and regions (Tan *et al.*, 2024). Managerial characteristics and stakeholder preferences further shape greenwashing (Pizzetti *et al.*, 2021), potentially

resulting in divergent evaluations of the same firm's behavior. Future research could investigate how internal firm characteristics interact with external regulatory environments to better explain greenwashing among listed firms.

Note

1. For instance, the *Overall Plan for Ecological Civilization System Reform*, issued by the Central Committee of the Communist Party of China and the State Council, outlines the phased introduction of mandatory environmental disclosure for listed firms. This has encouraged many firms to disclose their environmental information. While the policy is designed to strengthen environmental protection and has motivated many firms to disclose environmental information, it has also prompted some firms to respond with symbolic commitments and greenwashing, exaggerating their environmental performance (Xing *et al.*, 2021).
2. Social trust denotes individuals' general belief in the trustworthiness of others in society, reflecting a generalized tendency to trust, commonly referred to as "general trust." We measure regional social trust using data from the *China Entrepreneur Survey System*, with scores calculated as the weighted average of regional social trust rankings.
3. Media coverage data are drawn from the *Wisers Chinese News Database*: <https://www.wisers.com.cn>
4. China Greenwashing List (2023–2024), available at: <https://www.infzm.com/contents/276746>

Appendix

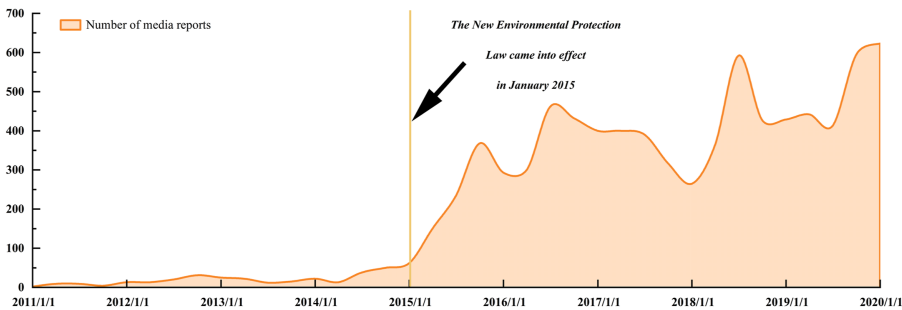


Figure A1. The number of media reports on corporate ESG. Source(s): Authors' own work

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